

CAPSTONE PROJECT · AWS CLOUD COMPUTING



Smart Garden Manager

Serverless IoT Monitoring Platform on AWS

Collect · Process · Store · Visualize · Alert

Yilin Xie · August 19, 2026



The Challenge

Many people struggle with plant care — they lack the data to know when soil is too dry or temperature too hot.

TRADITIONAL IoT

- Servers need to be managed
- Complex networking
- Scaling is time-consuming
- Pay for idle time – 24/7
- High operational overhead

SERVERLESS

- No servers to manage
- Auto-scaling
- Only pay for what is used
- Built-in high availability
- Focus on logic, not infrastructure

How do we solve the plant care problem without the complexity of traditional IoT?

Project objectives

01

Collect data

Temperature, humidity
and soil moisture of
sensors

02

**Process
automatically**

Validation and processing
without manual
intervention

03

Efficient storage

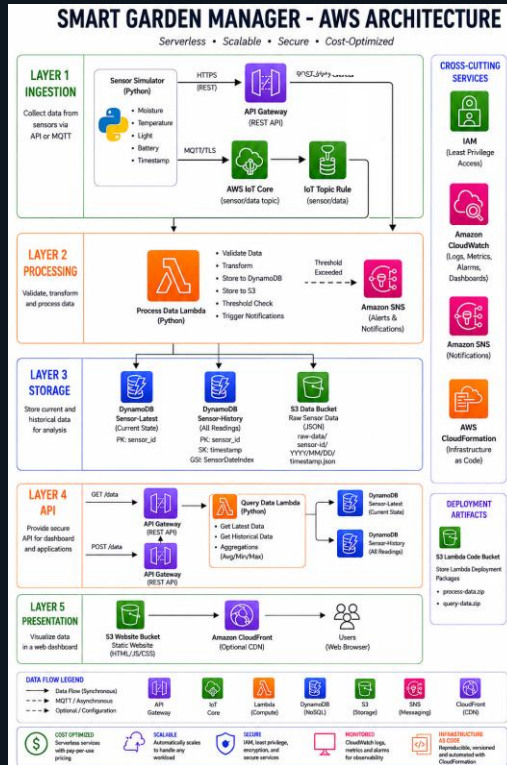
Current condition and
complete history

04

Monitor & Alert

Live dashboard and email
notifications

System Architecture



5 LAYERS

Layer 1 · Ingestion → API Gateway / IoT Core

Sensor simulator → API Gateway (HTTPS) or IoT Core (MQTT)

Layer 2 · Processing → Lambda

Lambda: validate, transform, store, threshold check, SNS

Layer 3 · Storage → DynamoDB / S3

DynamoDB latest & history · S3 raw JSON archive

Layer 4 · API → API Gateway

API Gateway → Query Lambda: latest, history, aggregations

Layer 5 · Presentation → S3 / CloudFront

S3 website bucket → CloudFront → users

Cross-cutting: IAM · CloudWatch · SNS · CloudFormation

AWS Services Used

10 SERVICES

IoT Core

MQTT Ingestion

API Gateway

HTTPS Ingestion

Lambda

Processing

DynamoDB

Data storage

S3

Website & Archive

CloudFront

Content Delivery

SNS

Email alerts

CloudWatch

Monitoring

CloudFormation

Infrastructure as
Code

IAM

Least Privilege
Security

Data Processing Pipeline and Dashboard

THE JOURNEY OF A SENSOR READING TO VISUALIZATION

STEPS 1-4 · INGESTION TO QUERY

1

Ingestion

Sensor simulator → API Gateway (HTTPS) or IoT Core (MQTT)

2

Processing

Process Lambda: validate, transform, threshold check, SNS

3

Storage

DynamoDB latest · DynamoDB history · S3 raw archive

4

Query

API Gateway → Query Lambda: latest, history, aggregations

STEP 5 · VISUALIZATION



4 live metrics · 3 charts · stats · data table

Alerting & Thresholds

SIX CHECKS ACROSS THREE METRICS

Metric	Low	High
Soil moisture	< 30 %	> 80 %
Temperature	< 5 °C	> 35 °C
Humidity	< 40 %	> 90 %

ALERT FLOW



Any breach triggers a single SNS e-mail containing the violated thresholds and all current readings.

Alert message example

SMART GARDEN ALERT

=====

Sensor: sensor-001

Time: 2026-08-17T15:40:51Z

ALERTS DETECTED

1. Low soil moisture: 25.8 % (threshold 30.0 %)
2. Low humidity: 36.7 % (threshold 40.0 %)

CURRENT VALUES

- Temperature: 25.8 °C
- Humidity: 36.7 %
- Soil moisture: 25.8 %

Security architecture

ANCHORED IN EVERY LAYER



IAM & Authentication

- Least privilege roles per service
- Certificate-based IoT authentication
- No passwords in the code



Data protection

- S3 with AES-256 encryption
- DynamoDB encrypted at rest
- Communication only via HTTPS/TLS



Access control

- S3 blocks public access
- Policies enforce HTTPS
- CloudFront Protects the Website Bucket

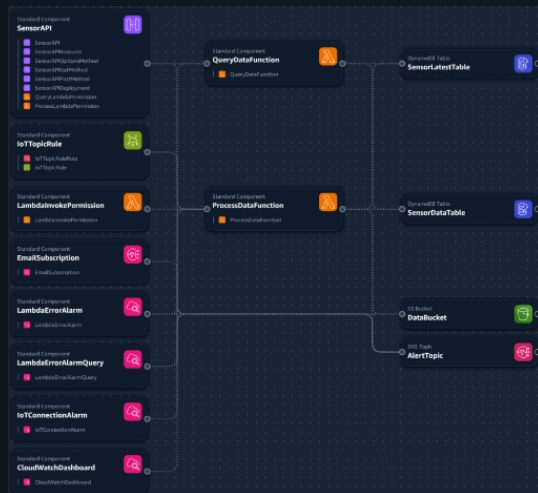
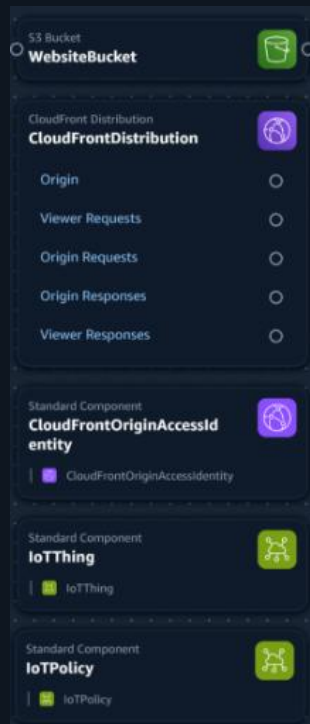


Monitoring & Recovery

- CloudWatch Logs as an Audit Trail
- Point-in-Time Recovery in DynamoDB
- S3 versioning and alarms

Infrastructure as Code

CLOUDFORMATION



3-5 mins

Deploy the entire infrastructure with
one command

One template defines everything

- S3 buckets
- DynamoDB Tables
- Lambda Functions
- API Gateway
- IoT Core
- All supporting resources

Repeatable deployments

Versioned infrastructure

Consistent environments

Easy rollback

Simulator Modes

3 OPERATING MODES

Offline

No AWS connection required

Ideal for development and testing
– at no cost

API

Sending via HTTPS to API
Gateway

Easy to set up, no certificates
required

MQTT

Connect to AWS IoT Core via
TLS

The Secure, Production-Ready IoT
Pattern

Important: API and MQTT modes use the same Process Lambda. Switching does not affect the storage or the dashboard.

Live Demo

TERMINAL · SIMULATOR OUTPUT

```
>> python src\simulator\sensor_simulator.py --mqtt --env --websocket --interval 5 --max-readings 10
```

```
...  
[2026-08-17 15:11:36] [INFO] MQTT OK topic=sensor/data reading_id=sensor-001-000001  
[2026-08-17 15:11:36] [INFO] READING 000001 | T: 25.1C | H: 39.4% | M: 46.4% | Batt: 92.0%  
[2026-08-17 15:11:42] [INFO] MQTT OK topic=sensor/data reading_id=sensor-001-000002  
[2026-08-17 15:11:42] [INFO] READING 000002 | T: 24.5C | H: 37.9% | M: 44.0% | Batt: 94.8%  
...  
[2026-08-17 15:12:23] [INFO] MQTT OK topic=sensor/data reading_id=sensor-001-000010  
[2026-08-17 15:12:23] [INFO] READING 000010 | T: 25.5C | H: 40.6% | M: 37.8% | Batt: 95.6%  
[2026-08-17 15:12:23] [INFO] Disconnected from AWS IoT Core.  
[2026-08-17 15:12:23] [INFO] =====  
[2026-08-17 15:12:23] [INFO] MQTT SIMULATION STATISTICS
```

DASHBOARD · LIVE UPDATES



Alert flow: sensor reading → threshold check → SNS publish → e-mail alert

1 Start Simulator

2 Watch Terminal

3 Show dashboard

4 Trigger the alarm

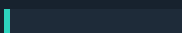
Results & Cost Overview

FUNCTIONALLY COMPLETE · ROUGHLY \$2.43 PER MONTH

FUNCTIONALLY ACHIEVED

- Complete IoT data lifecycle
- Automated processing with Lambda
- Storage in DynamoDB and S3
- Live dashboard with visualization
- Infrastructure as Code

MONTHLY COST BY SERVICE (USD)

IoT Core (MQTT)		\$0.68
Amazon S3		\$0.50
Amazon SNS		\$0.50
CloudWatch		\$0.50
DynamoDB		\$0.25
Lambda (Free Tier)		\$0.00

COMPARISON

Traditional IoT Platform

\$50 – \$200

Self-hosted EC2

\$30 – \$100

Smart Garden Manager

~\$2.43

Lowest cost of the three options

Usage basis: 518,400 MQTT messages/month · 5 GB S3 storage · 1,000 notifications

Outlook & Further Development

ROADMAP

In the short term

- Connect real physical sensors
- Advanced analytics with Athena on S3
- Cognito for secure authentication
- Dashboard fully mobile-optimized

Medium-term

- Machine Learning for Anomaly Detection
- Several sensors in the garden
- Predictive analytics
- Intelligent irrigation control

Long-term

- Multi-garden platform
- Smart home integration
- Advanced analytics dashboard
- Self-learning threshold values

From prototype to production-ready system.

Questions & Discussion

CONTACT

Presenter

Yilin Xie

GitHub

github.com/xieyilin77/Smart-Garden

LinkedIn

linkedin.com/in/yilin-xie6

LIKELY QUESTIONS

Why serverless over traditional servers?

No server management, automatic scaling, pay-per-use

How does it scale to thousands of sensors?

DynamoDB scales automatically, Lambda runs in parallel

What about cold starts?

Mitigated with provisioned concurrency

How secure is this solution?

Five layers: TLS, certificates, IAM, encryption, monitoring

What is the monthly cost at scale?

About \$15–20 for 50 sensors (~\$0.30–0.50 per sensor)

Thank you for your attention — happy to discuss the project, the architecture, or the implementation details.